

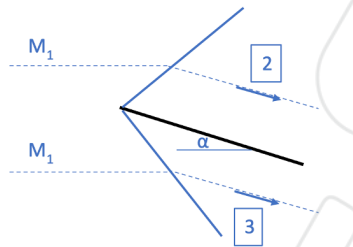
AERO50001 Aerodynamics 2 Problem Set #7

Extra practice: Linear supersonic theory

7.1 Flat plate

★★☆ 8-15 mins

Use Ackeret's theory to estimate the pressure coefficient on each side of a flat plate in a flow at $M_1 = 2.5$ at an incidence angle of 3° .



- (a) At the top of the plate, C_{p2} .
- (b) At the bottom of the plate, C_{p3}

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$$(a) \quad C_{p2} = \frac{2(-\theta)}{\sqrt{M_1^2 - 1}} = -0.0457$$

$$(b) \quad C_{p3} = -\frac{2\theta}{\sqrt{M_1^2 - 1}} = +0.0457$$